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MILITARY SPECIFICATION

SCAFFOLDING COMPONENTS, STEEL (PIPE, TUBE, AND COUPLER)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel pipe, tube, and coupler scaffolding components including pipe, tubing, couplers, and associated fittings.

1.2 Classification. Pipe, tube and coupler scaffolding components shall be of the following components, sizes, lengths, types, and styles as specified (see 6.2 and 6.5):

Component I - Right angle coupler

Size 2 by 2-inch.
Size 2-1/2 by 2-inch.
Size 2-1/2 by 2-1/2-inch.

Component II - Swivel coupler.

Size 2 by 2 inch.
Size 2-1/2 by 2-inch.
Size 2-1/2 by 2-1/2-inch.

Component III - Twist lock fitting.

Size 2-inch tubing.
Size 2-inch pipe.
Size 2-1/2-inch pipe.

Type - male.
Type - female.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Component IV - Pipe and tubing.

Size 2-inch tubing.
Size 2-inch pipe.
Size 2-1/2-inch pipe.

Length - 6 feet.
Length - 8 feet.
Length - 10 feet.
Length - 13 feet.

Type - with twist lock fitting.
Type - without twist lock fitting.

Component V - Floor plate.

Size 2-inch tubing.
Size 2-inch pipe.
Size 2-1/2-inch pipe.

Type - with twist lock fitting.
Type - with female cylinder.

Component VI - Caster.

Size 2-inch tubing.
Size 2-inch pipe.
Size 2-1/2-inch pipe.

Type - with twist lock fitting.
Type - with female cylinder.

Style - with rubber wheels.
Style - with steel wheels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ISO/R261 - ISO Metric Screw Threads, General Plan, 0.25 to 300 mm.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel. (DoD adopted)
A 47 - Malleable Iron Castings. (DoD adopted)
A 120 - Pipe Steel Black and Hot-Dipped Zinc Coated (Galvanized) Welded and Seamless, for Ordinary Uses. (DoD adopted)
A 123 - Zinc (Hot Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars and Strip. (DoD adopted)
A 153 - Zinc Coating (Hot Dip) on Iron and Steel Hardware. (DoD adopted)
A 370, Sup III - Mechanical Testing of Steel Products, Methods and Definitions for. (DoD adopted)
A 449 - Quenched and Tempered Steel Bolts and Studs. (DoD adopted)
A 501 - Hot-Formed Welded and Seamless Carbon Steel Structural Tubing.
A 563 - Carbon and Alloy Steel Nuts. (DoD adopted)
A 668 - Steel Forgings, Carbon and Alloy, for General Industrial Use. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM FREIGHT CLASSIFICATION RATINGS, RULES AND REGULATIONS
Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Agent Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Pipe, tube,and coupler scaffolding components are the fundamental building units of engineered scaffold structures (see 6.1).

3.2 First article. When specified (see 6.2), scaffolding components of the type, style, length, and size shall be subjected to first article inspection (see 4.3 and 6.4).

3.3 Standard commercial product. Each scaffolding component of the same classification shall, as a minimum, be in accordance with the requirements specified herein and may be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeable items. All scaffolding components of the same classification, furnished under any specific contract, shall be physically and mechanically interchangeable. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable.

3.6 Right angle coupler. The right angle coupler shall have two drop forged caps, a drop forged one piece body, and a securing bolt for each cap. Forgings shall conform to ASTM A 668. Body and caps shall be

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contoured to fit the outside diameter of the applicable pipe or tubing size. Coupler bodies shall be 4 inches ± 1/4 inch wide for 2-inch pipe or tubing and 4-1/2 inches ± 1/4 inch wide for 2-1/2-inch pipe. Each cap shall hinge on an annealed steel pin and shall be provided with an opening to receive the bolt without removing the nut. The cap shall be designed so as to swing from fully closed to fully open in an approximate arc of 90 degrees. A hinge stop shall be incorporated to maintain this 90 degree angle for the fully open position. The cap shall also be provided with a minimum 1/16-inch lip to prevent the bolt from slipping out of the opening after tightening. The two bolts shall meet the requirements of 3.13.2.1 and shall be either 1/2-13 NC-2A or M14 captive bolts capable of movement only in a radial plane into and out of the cap opening for assembly or disassembly. Nuts shall meet the requirements of 3.13.2.2 and shall be 1/2-13 NC-2B or M14 heavy hex single chamfered or collar nuts of suitable dimensions to permit use of a 7/8-inch wrench for tightening. M14 nuts shall be used only with M14 bolts. The 1/2-13 NC-2B nuts shall be used only with 1/2-13 NC-2A bolts. Thread length of the bolt shall be sufficient to insure that three or more unused threads shall remain visible after the bolt has been torqued to the maximum test torque specified in 4.8.1.4. The bolt end threads on assembled couplers shall be upset to discourage removal of the nut for field replacement and to prevent loss of the nut during disassembly. Unless otherwise specified (see 6.2), the coupler shall be of right-handed configuration as defined in 6.7. The bolt-nut-coupler assembly shall withstand a torque of 125 foot-pounds (ft-lb) in a nonlubricated condition. The coupler shall be furnished as an assembled unit. The unit shall have a minimum frictional holding capacity of 2,000 pounds (lb), and shall support a load of 6,000 lb without rotating in excess of 10 degrees (see figure 1). Unless otherwise specified (see 6.2), the unit shall withstand a minimum of 16,000 lb tensile pull. The coupler assembly shall be finished in accordance with 3.12.

3.7 Swivel coupler. The swivel coupler shall conform to the requirements of 3.6 except that the body shall be provided in two pieces joined by the manufacturer's standard connection. The connection shall allow turning of the coupler to any desired angle. The unit shall have a minimum frictional holding capacity of 2,000 lb. The unit shall support a load of 3,000 lb without rotating in excess of 10 degrees and shall withstand a minimum of 7,000 lb of tensile pull.

3.8 Twist lock fittings. Twist lock fittings shall consist of a male fitting and a female fitting. The female fittings shall either be malleable iron castings conforming to ASTM A 47 or steel forgings conforming to ASTM A 668. The male fittings shall be steel forgings conforming to ASTM A 668. The male fitting shall be nominal 4 inches in length with a minimum of 2-inch length available to be press fit into the tube or pipe and then be attached to the tube or pipe by the manufacturer's standard method of attachment. The twist locking portion shall be nominal 15/16-inch diameter with two tapered, locking lugs of the manufacturer's standard configuration. Female fitting shall have a minimum of 2-inch length to be press fit into the tube or pipe and then be attached to the tube or pipe by the manufacturer's standard method of attachment. The fitting shall accept the male twist locking portion and lock with a 90 degree rotation. The maximum outside diameter of the coupler shall conform to the outside diameter of the pipe or tube, so that the joint is smooth

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and flush. When installed in a pipe or tubing, the twist lock fitting-pipe or tube assembly shall withstand a tensile force of 4,000 lb and a compressive force of 20,000 lb. The twist lock fitting shall be finished in accordance with 3.12.

3.9 Pipe and tubing. Tubing shall be high strength, structural steel tubing and shall be 1.90-inch outside diameter and 0.095 inch minimum wall thickness, for 2-inch designated size. Minimum yield strength shall be 50,000 pounds per square inch (lb/in^2) and ultimate tensile strength shall be 75,000 lb/in^2. Pipe shall conform to ASTM A 501 as referenced in ASTM A 36 and shall be 1.90-inch outside diameter and 0.145-inch minimum wall thickness for 2-inch designated size (ASTM A 501 1-1/2 inch nominal size) and 2.375-inch outside diameter and 0.154-inch minimum wall thickness for 2-1/2-inch designated size (ASTM A 501 2-inch nominal size). Tubing and pipe shall be either 2-inch tubing, 2-inch pipe, or 2-1/2-inch pipe, as specified (see 6.2); with or without a female twist lock fitting conforming to 3.8 attached to one end and a male twist lock fitting conforming to 3.8 attached to the other end, as specified (see 6.2), and in the length specified (see 6.2). Pipe and tubing shall be finished in accordance with 3.12.

3.10 Floor plate. The floor plate shall be either a malleable iron casting conforming to ASTM A 47 or a weldment made up of a steel plate and a suitable fitting or cylinder. The plate shall be nominal 8-inch diameter or 6-inch by 6-inch square of 0.25 inch minimum thickness for four equally spaced 9/16-inch screw holes near the perimeter for mounting purposes. The base shall be furnished with either a twist lock fitting conforming to 3.8 or with a minimum 4-inch length cylinder or steel bar stock sized to accept the outside diameter of the applicable pipe or tubing as specified (see 6.2). When the floor plate, fitted with a cylinder, is specified, the cylinder shall be fitted with a thumb screw to lock the floor plate assembly to the scaffold member. The plate shall withstand a shear force of 5,000 lb and a compressive force of 20,000 lb. The floor plate shall be finished in accordance with 3.12.

3.11 Caster. Casters shall be nominal 8-inch diameter with steel or rubber wheels as specified (see 6.2). Caster shall be provided with a foot actuated locking device. Caster shall either be provided with a male twist lock fitting conforming to 3.8, or with a minimum 4-inch length cylinder, sized to accept the outside diameter of the applicable pipe or tubing as specified (see 6.2). When the caster, fitted with the cylinder, is specified, the cylinder shall be provided with a thumb screw to lock the caster assembly to the scaffold member. The steel wheel assembly shall support a load of 8,000 lb. The rubber wheel caster assembly shall support a load of 500 lb. The caster shall be clearly marked with the safe working load as assembled. The caster assembly, except for the rubber wheels shall be finished in accordance with 3.12.

3.12 Finish. Scaffolding components to be finished, other than steel tubing or pipe shall be finished in accordance with ASTM A 153. Finish of structural steel tubing, after cutting and seam welding of the tube, shall conform to ASTM A 123. Finish of steel pipe shall conform to ASTM A 120.

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3.13 Workmanship.

3.13.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.13.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed.

3.13.2.1 Bolts. Bolts shall conform to ASTM A 449 for quenched and tempered medium carbon steel bolts with a minimum tensile strength of 120,000 lb/in^2. Bolts shall be commercially smooth and free from burrs, laps, seams, cracks or other defects which would make them unsuitable for the intended application. Metric threads shall conform to ANSI ISO/R261.

3.13.2.2 Nuts. Nuts shall conform to ASTM A 563 for heavy hex nuts for use with quenched and tempered medium carbon steel bolts. Metric thread shall conform to ANSI ISO/R261.

3.13.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed.

3.13.3.1 Rivets. Rivets shall conform to industry standard for carbon steel rivets. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value. Rivets shall be commercially smooth and free from cracks (burst), duds or other defects which would make them unsuitable for the intended application.

3.13.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with AWS D1.1. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.13.4.1 Welder qualification. All welders shall be qualified for the types of welds and for welding positions required for use in the fabrication of each unit. Qualification shall be on the type of base material and filler utilized in actual construction or on similar alternate materials not requiring operator requalification. Welder qualification shall be performed in accordance with the applicable sections of AWS D1.1. Copies of the qualification record for each qualified welder shall be kept by the manufacturer or contractor and shall be available to authorized government inspectors.

3.13.5 Castings. Castings shall be ferritic malleable iron conforming to ASTM A 47 malleable iron castings. No repairing, plugging, or welding of any kind shall be permitted. Castings surface shall be

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smooth and free from scale and burned on sand. Castings shall be sound and free of misplaced coring, warping or any other defect which reduces the ability of the casting to perform the intended function.

3.13.6 Forgings. Forgings shall conform to ASTM A 668 for steel forgings for general industrial use. Forgings shall be sound and free from cold shuts, laps, folds, chops, or any other defect which reduces the ability of the forging to perform the intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification (see 6.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Packaging inspection (see 4.10).

4.3 First article inspection. First article inspection shall be performed on one complete scaffolding component of each type furnished when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examinatin of 4.7 and the applicable tests of 4.8. All tests required for a component of a particular classification shall be performed on a single test unit in the order in which they are specified in 4.8. The first article may be a standard production item from the contractor's current inventory, provided the scaffolding component meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining scaffolding components of that type to be furnished under the contract.

4.3.1 First article inspection report. The contractor shall furnish a first article inspection report in accordance with the data ordering document specified in 6.3.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on the sample scaffolding components selected in accordance with 4.6. This inspection shall include the examination of 4.7 and the tests of 4.8, which are applicable to the component being inspected.

4.4.1 Certification. A certificate of compliance shall be prepared certifying that the materials used for the couplers, bolts, nuts, pipe, tubing, and accessories conform to the specified requirements of 3.6 through 3.12 (see 6.3).

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4.5 Inspection lot. All components from a single manufacturer of the same size, type, length, and style as applicable, offered to the Government at one time, shall be considered a lot for purpose of inspection.

4.6 Sampling. A random sample of components shall be selected from each lot in accordance with MIL-STD-105.

4.6.1 Sampling for examination. Examination of the components shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.6.2 Sampling for tests. Tests of the components shall be based on inspection level S-2 and an AQL of 4.0 percent defective.

4.7 Examination. Each sample selected in accordance with 4.6.1 shall be examined for compliance with the requirements specified herein. Any modification of the contractor's standard product to comply with specified requirements, or any necessary modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Examination shall be conducted as specified in table I.

TABLE I. List of defects.

Paragraph	Defect
3.4	Material not as specified, obviously defective, damaged or not suitable for the intended purpose.
3.5 through 3.11	Components of the same classification not physically and mechanically interchangeable as specified. Dimensions not within tolerances specified.
3.12	Finish not as specified.
3.13 through 3.13.4	Workmanship does not meet the requirements and standards specified.

4.8 Tests. Each sample selected in accordance with 4.6.2 shall be tested to determine compliance with the requirements specified herein. Tests shall be conducted on each sample selected and shall be conducted in the order set forth in the applicable test section for the sample being tested. Any visible or measureable permanent deformation in the test sample, or failure of the sample to meet any applicable test requirement, shall constitute failure of the sample.

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4.8.1 Right angle coupler tests.

4.8.1.1 Test for rotation under load. The coupler, connecting two tubes at right angles and restrained from slipping on the vertical tube (see figure 1), shall be subjected to a load of 6,000 lb applied to the horizontal tube. During this test the coupler shall not rotate through an angle exceeding 10 degrees from the horizontal. Each gripping surface of the coupler shall be so tested in both directions. Coupler nuts shall be torqued to 50 ± 5 ft-lb for this test.

4.8.1.2 Test for slip. The coupler, connecting two tubes at right angles shall be subjected to a load of 2,000 lb applied to the horizontal tube, as shown in figure 2, with less than 0.25 inch vertical slip. During this test, the coupler shall not rotate through an angle exceeding 10 degrees from the horizontal. Each gripping surface of the coupler shall be so tested in both directions. Coupler nuts shall be torqued to 50 ± 5 ft-lb for this test.

4.8.1.3 Test for tensile strength. Unless otherwise specified (see 3.6 and 6.2), the coupler, connecting two tubes at right angles, shall be subjected to a minimum load of 16,000 lb in tension, as shown in figure 3, without failure of the coupler. Coupler nuts shall be torqued to 50 ± 5 ft-lb for this test.

4.8.1.4 Torque test. Each section of the coupler shall be clamped to a tube and the coupler-bolt-nut assembly torqued to a value of 125 ft-lb in a dry nonlubricated condition. During this test the coupler-bolt-nut assembly shall undergo no permanent deformation.

4.8.2 Swivel coupler tests.

4.8.2.1 Test for rotation under load. The coupler, connecting two tubes at right angles and prevented from slipping on both tubes (see figure 1) shall be subjected to a load of 3,000 lb. During this test the coupler shall not rotate through an angle exceeding 10 degrees from the horizontal. Each gripping surface of the coupler shall be so tested in both directions. Coupler nuts shall be torqued 50 ± 5 ft-lb for this test.

4.8.2.2 Test for slip. The coupler, connecting two tubes at right angles, shall be subjected to a load of 2,000 lb applied to the horizontal tube, as shown in figure 2, with less than 0.25-inch vertical slip. During this test, the coupler shall not rotate through an angle exceeding 10 degrees from the horizontal. Each gripping surface of the coupler shall be so tested in both directions. Coupler nuts shall be torqued 50 ± 5 ft-lb for this test.

4.8.2.3 Test for tensile strength. The coupler connecting two tubes at right angles, shall be subjected to a load of 7,000 lb in tension, as shown in figure 3, without failure of the coupler. Coupler nuts shall be torqued to 50 ± 5 ft-lb for this test.

4.8.2.4 Torque test. The coupler shall be tested in accordance with 4.8.1.4.

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4.8.3 Twist lock fitting tests.

4.8.3.1 Test for tensile strength. A twist lock fitting assembly connecting two tubes (see figure 4), shall be subjected to a tensile load of 4,000 lb with no more than 0.25 inch slip and with no failure of the fitting.

4.8.3.2 Test for compressive strength. A twist lock fitting assembly connecting two tubes (see figure 5), shall be subjected to a compressive load of 20,000 lb with no permanent deformation or failure of the fitting.

4.8.4 Floor plate tests.

4.8.4.1 Test for shear strength. The floor plate (see figure 6), shall be subjected to a shear load of 5,000 lb without failure of the floor plate.

4.8.4.2 Test for compressive strength. The floor plate (see figure 7), shall be subjected to a compressive load of 20,000 lb without failure of the floor plate.

4.8.5 Caster tests.

4.8.5.1 Test for compression, rubber wheel caster. The caster (see figure 8), shall be subjected to a load of 500 lb without failure or permanent deformation.

4.8.5.2 Test for compression, steel wheel caster. The caster (see figure 8), shall be subjected to a load of 8,000 lb without failure or permanent deformation.

4.9 Bolt tensile testing. Bolts shall be full size tensile tested to insure conformance to the tensile requirements (see 3.13.2.1). Tensile testing shall be done in accordance with supplement III of ASTM A 370. Manufacturer's certified test report for the last completed set of mechanical tests for the size of bolt shall be acceptable as indication of compliance with the tensile requirement for the bolt.

4.10 Packaging inspection. The preservation-packaging, packing, and marking of the scaffolding components shall be inspected to verify conformance to the requirements as specified herein.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Coupler, fittings, joint pins, and casters of like description shall be packaged in close-fitting boxes conforming to PPP-B-636, class weather resistant. The contents within the boxes shall be cushioned to prevent movement. The boxes shall be closed and sealed in accordance with method IV as specified in PPP-B-636. The floor plates shall be wired together with soft annealed wire in quantities of four each or as recommended by the contractor.

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5.1.2 Level C. The scaffolding components shall be packaged in accordance with the standard practice of the contractor.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pipe and tubing. Unless otherwise specified (see 6.2), pipe and tubing of the same size and length shall be bundled in secured lifts not to exceed approximately 500 lb to form a nonshifting bundle. The pipe or tubing shall be arranged and secured with steel strapping. Nominal 2- by 4-inch wood battens shall be placed beneath all straps on top, bottom, and sides of the bundle. Strapping shall be 0.025 by 3/4 inch conforming to QQ-S-781, class 1, type I or IV, finish B and shall be stapled to the battens. A minimum of three straps shall be provided on each bundle. The center strap shall be spaced not to exceed 12 inches from the center of the bundle. The end straps shall be placed not more than 18 inches from each end of the bundle.

5.2.1.2 Other components. The packaged couplers, fittings, pins, casters, and floor plates shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The contents shall be cushioned, blocked, and braced to prevent movement.

5.2.2 Level B. The scaffolding components shall be packed as specified for level A, except the boxes shall conform to PPP-B-601, domestic type, or PPP-B-621, class 1.

5.2.3 Level C. The scaffolding components shall be packed in a manner which will ensure arrival at destination in satisfactory conditon and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classifications Rules and Regulations or National Motor Freight Classification.

5.3 Marking. Interior packages, snipping containers, and bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Steel 2-inch and 2-1/2-inch tube and coupler scaffolding components are inteded for use in engineered scaffold structures. Scaffold structures, using these components, may be defined as light duty, medium duty, or heavy duty, dependent on the desired load capacity and the resultant required post spacing as defined in CFR 1926.451 and ANSI 10.8. Scaffold structures to be erected utilizing the components specified herein should conform to the requirements of CFR 1926.451. ANSI 10.8, the SSI Recommended Scaffolding Erection Procedure and Scaffolding Safety Rules in addition to applicable local engineering guidelines. The components specified herein are not intended for use in construction of 3-1/2-inch outside diameter tube and coupler scaffolds.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Component, size, length, type, and styles required (see 1.2).
(c) When a first article is required for inspection and approval (see 3.2, 4.3, and 6.4).
(d) When right angle couplers shall be other than right-handed configuration (see 3.6).
(e) When right angle couplers shall be tension tested at other than 16,000 lb (see 3.6 and 4.8.1.3).
(f) Whether pipe or tubing shall be furnished with twist lock fittings installed and length required (see 3.9).
(g) Whether steel or rubber wheels are required (see 3.11).
(h) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).
(i) If level A packing of pipe and tubing is other than specified (see 5.2.1.1).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirements title	Applicable DID No.
4.1.1, 4.4.1, and 4.9	Certificates of Compliance	DI-E-2121
4.3.1	Reports, Test	DI-T-2072

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisiton functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4.1 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which

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has been previously acquired or tested by the Government, and that bidder offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Size designations. Designated sizes for scaffolding pipe and tubing vary from the standard nominal sizes as follows: 2-inch tubing and 2-inch pipe are 1.9-inch outside diameter and 2-1/2-inch pipe is 2.375-inch outside diameter. Other component designated sizes correspond to these actual dimensions.

6.6 Accessories. There are many additional scaffolding components available commercially. These items were not included in this specification since it was not the purpose of the specification to generate an exhaustive list of scaffolding components and accessories. If using activities find items not included, for which repeated procurements are required, they are requested to forward the attached DD 1426 to the preparing activity for consideration for inclusion of the items in a subsequent revision of this document.

6.7 Right-handed coupler configuration. The right-handed coupler configuration is defined with the following example: When a worker is using the left hand to install one half of the coupler on a horizontal pipe from underneath (horizontal hinge pin on bottom) with the clamping nut facing the installer, the clamping nut on the vertical half must be on the right hand side of the vertical pipe readily accessible to the right hand of the installer.

6.8 Changes from previous issue. Asterisks are not used in this revison to identify changes with respect to the previous issue due to the extensiveness of the changes.

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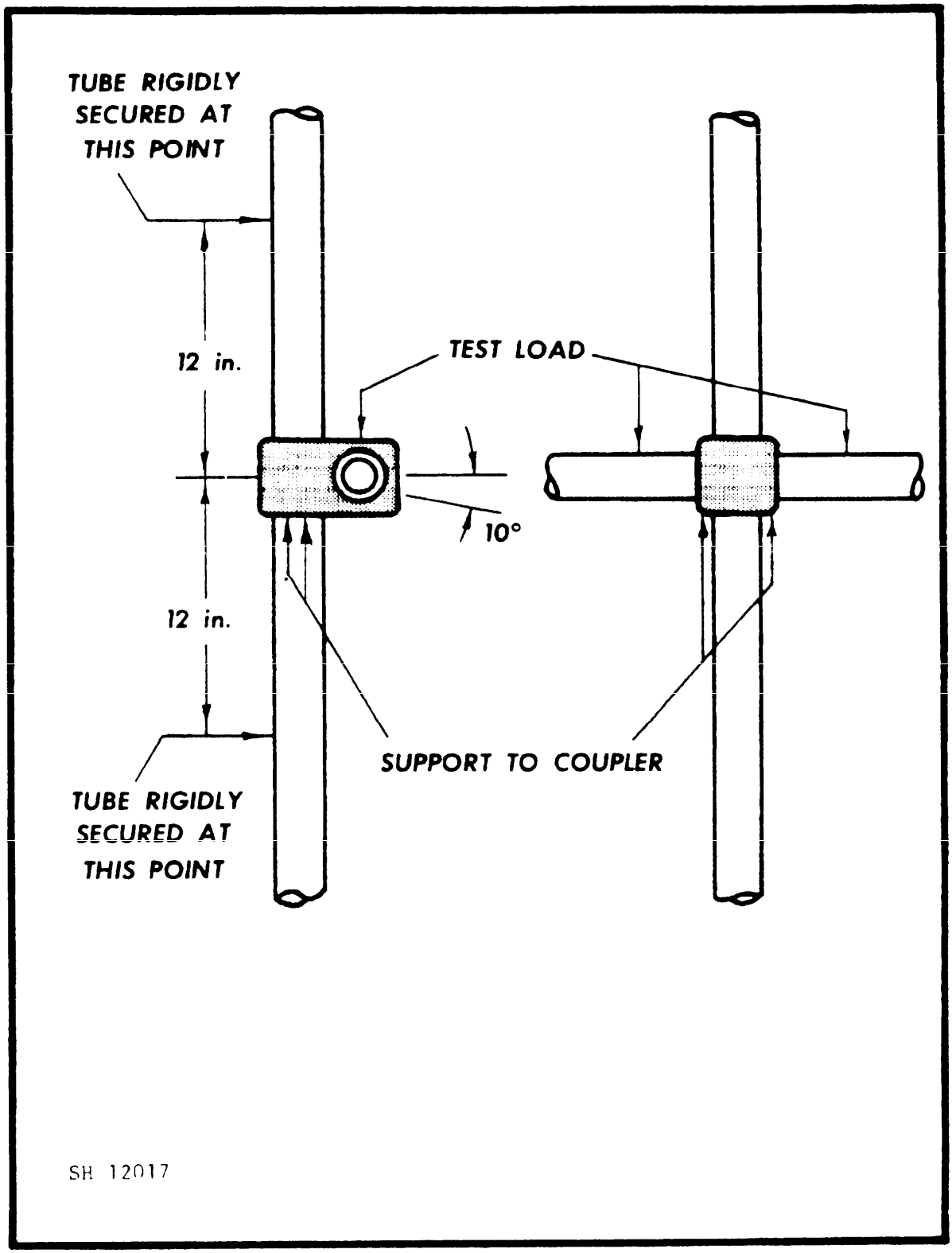


FIGURE 1. Test for distortion of right angle or swivel coupler.

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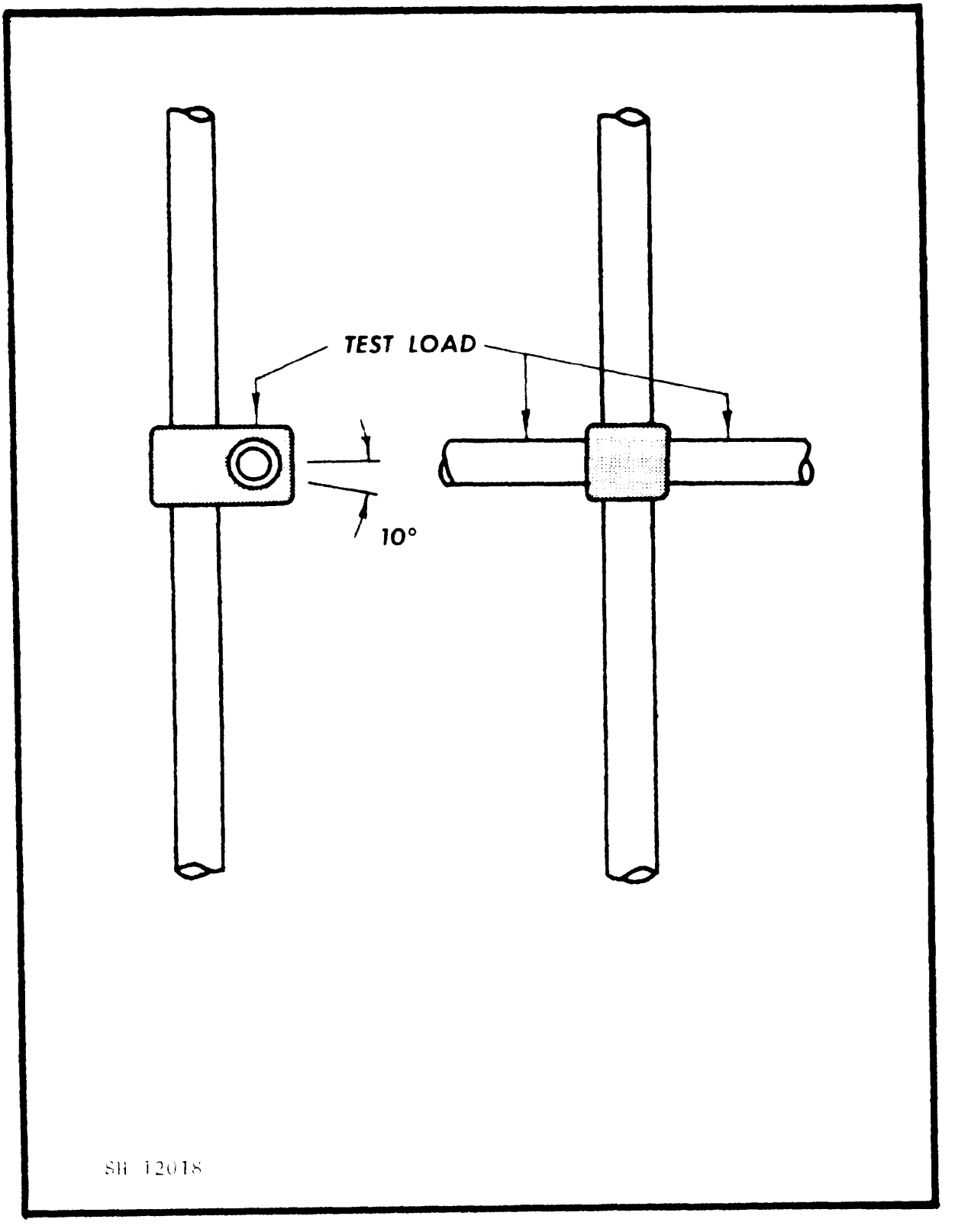


FIGURE 2. Test for slip of right angle or swivel coupler.

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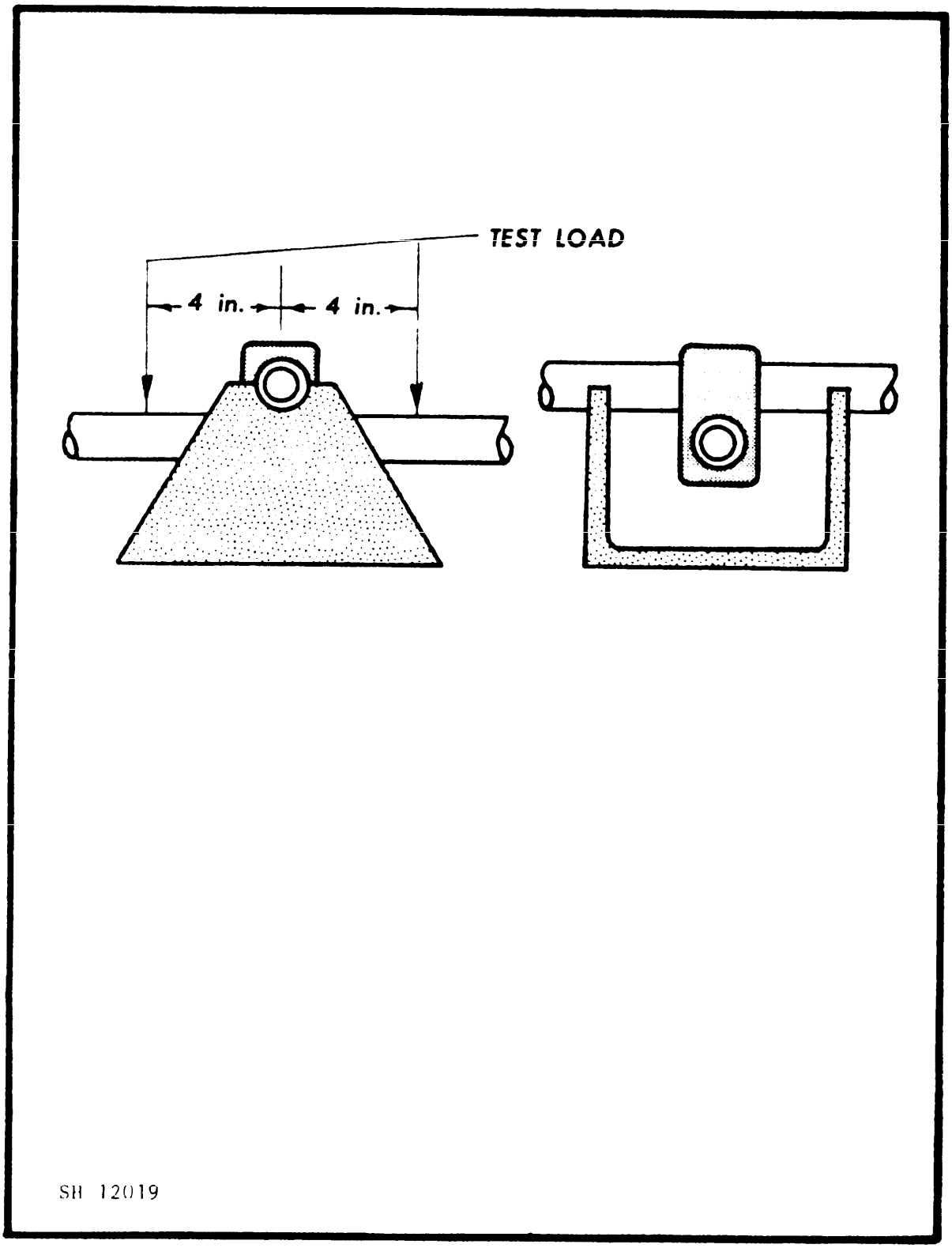


FIGURE 3. Test for tensile strength of right angle or swivel coupler.

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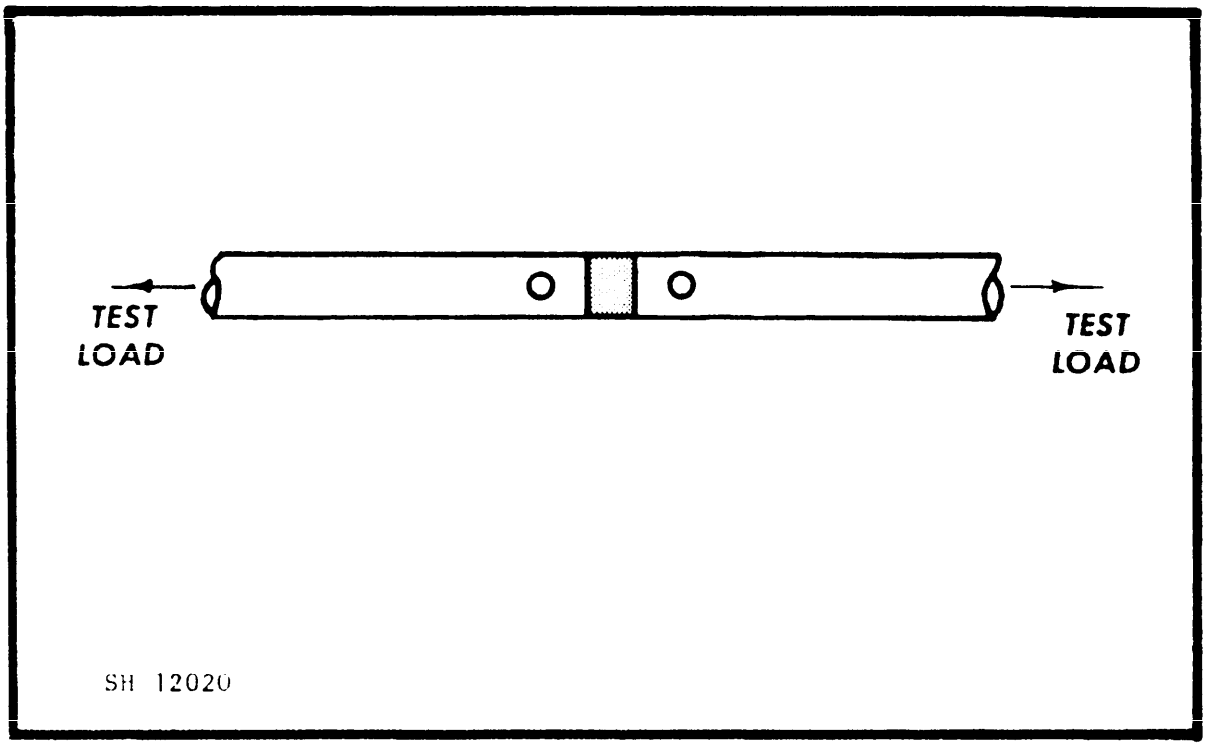


FIGURE 4. Test for tensile strength of twist lock couplers.

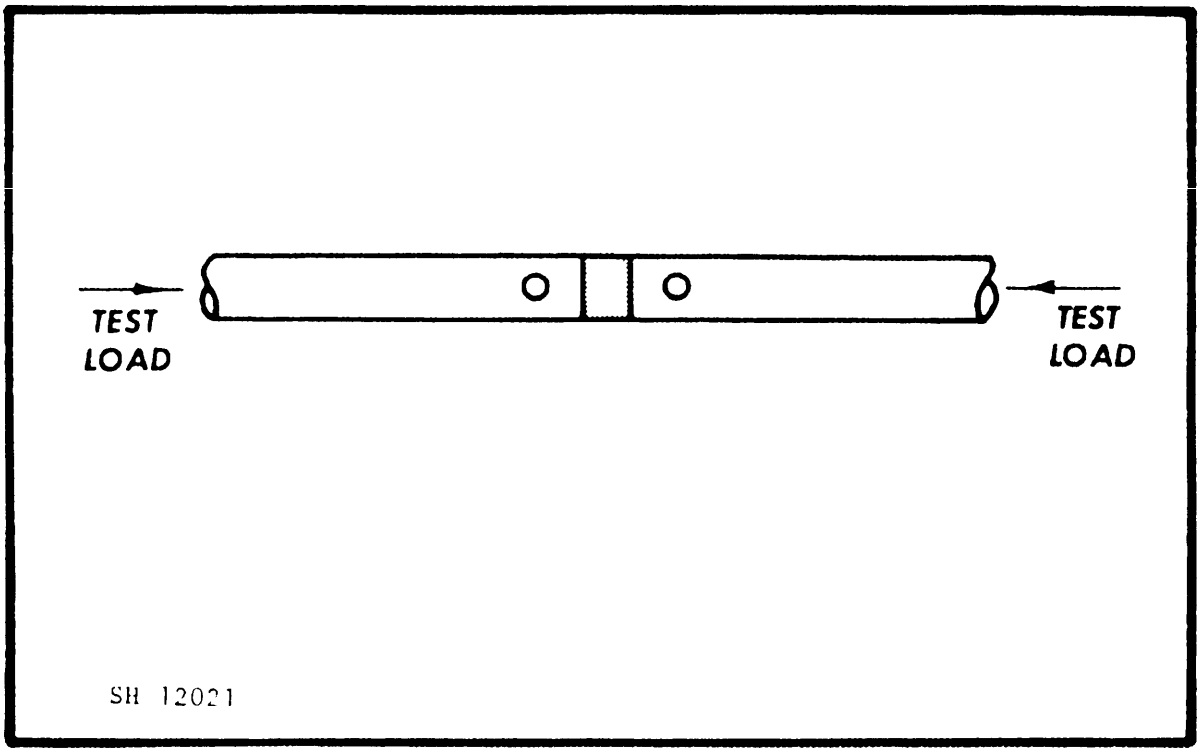


FIGURE 5. Test for compressive strength of twist lock couplers.

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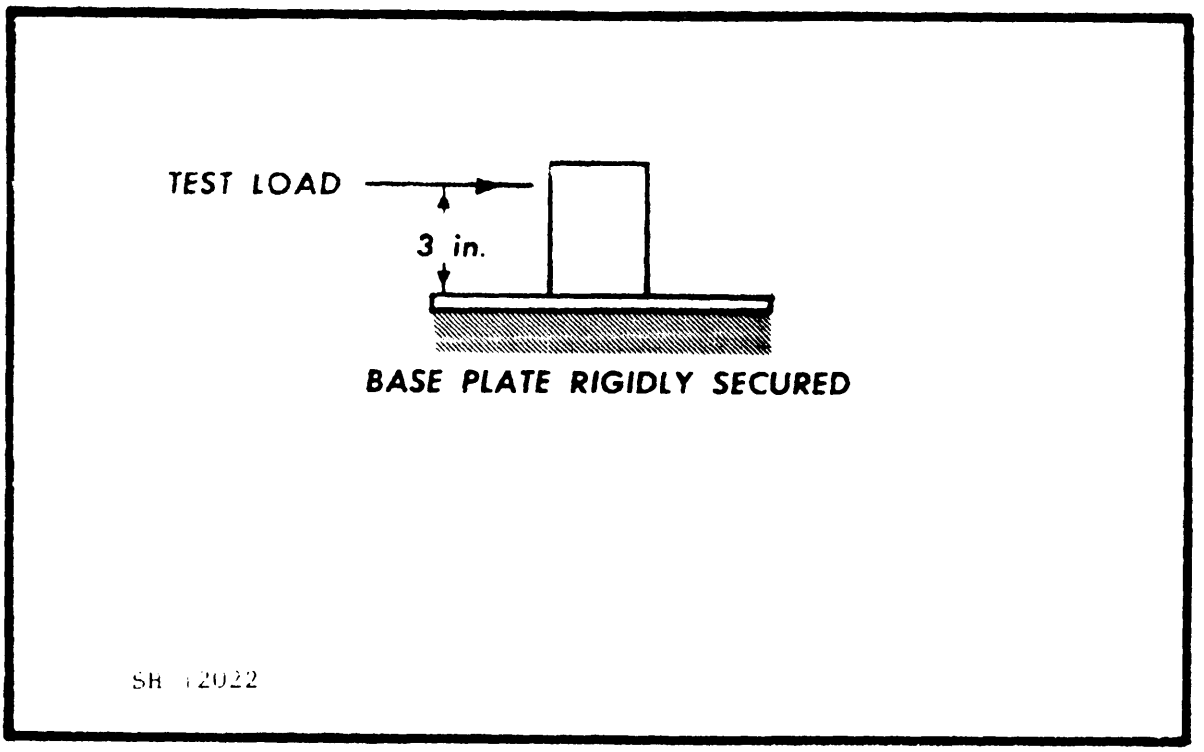


FIGURE 6. Test for shear of floor plate.

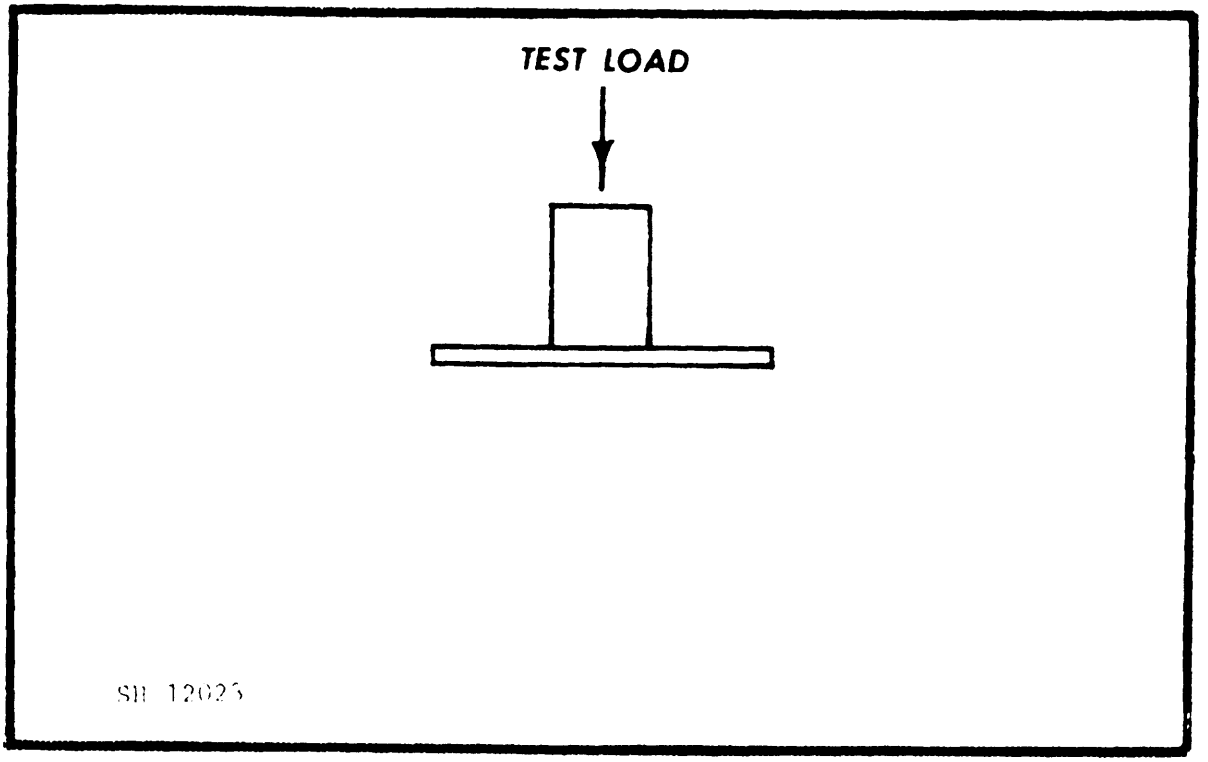
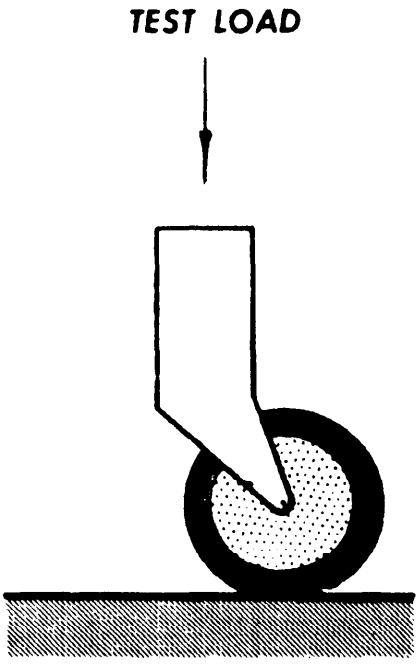


FIGURE 7. Test for compressive strength of floor plate.

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FIGURE 8. Test for compressive strength of caster.

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